Ores

The different types of ores are depicted in Figure 6.1.

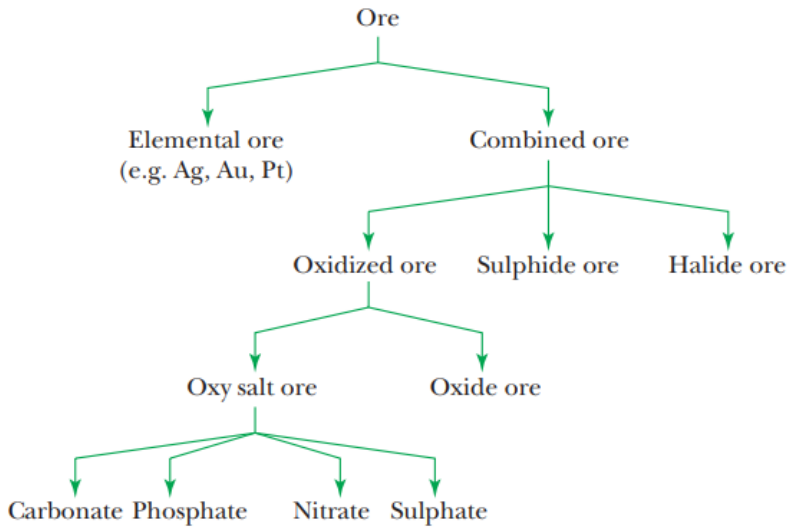


Figure 6.1 Different types of ores.

Some important ores of different types and their formulas are listed as follows.

1. **Oxide ores**

ZnO	Zincite	$CaMg_3(SiO_3)_4$	Asbestos
Fe_2O_3	Haematite	$MgO \cdot Al_2O_3$	Spinel
Fe_3O_4	Magnetite	$Al_2O_3 \cdot 3H_2O$	Gibbsite
$Al_2O_3 \cdot 2H_2O$	Bauxite	$Al_2O_3 \cdot H_2O$	Diaspore
$Fe_2O_3 \cdot 3H_2O$	Limonite	Al_2O_3	Corrundum
Cu_2O	Cuprite or Ruby copper	$PbCrO_4$	Crocoisite
MnO_2	Pyrolusite	$ZnO \cdot Fe_2O_3$	Franklinite
SnO_2	Tinstone or Casseterite	$ZnSiO_3$	Willemite
TiO_2	Rutile	$ZnSiO_3 \cdot ZnO \cdot H_2O$	Electric calamine
$FeCr_2O_4$ ($FeO + Cr_2O_3$)	Chromite ore	$K_2O \cdot Al_2O_3 \cdot 6SiO_2$	Feldspar
$Na_2B_4O_7 \cdot 10H_2O$	Borax or Tincal	$Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$	China clay or Kaolin
$Ca_2B_6O_{11} \cdot 5H_2O$	Colemanite		
U_3O_8	Pitch blende		
$FeO \cdot TiO_2$	Ilmenite		

2. **Sulphide ores**

PbS	Galena	Ag_2S	Argentite or silver glance
HgS	Cinnabar	$3Ag_2S \cdot Sb_2S_3$	Ruby silver
ZnS	Zinc blende or Sphalerite		
Cu_2S	Copper glance or Chalcocite		
$CuFeS_2$	Copper pyrite or Chalcopyrite		
FeS_2	Iron pyrite or Fool's gold		
Ag_2S	Silver glance or Argentite		

3. Halide ores

$NaCl$	Rock Salt
KCl	Sylvine
CaF_2	Fluorspar

Na_3AlF_6	Cryolite
$AgCl$	Horn silver
$KCl \cdot MgCl_2 \cdot 6H_2O$	Carnalite
$Cu_2Cl(OH)_3$	Atacamite

4. Oxy salt ores

a. Carbonate ores

$CaCO_3$	Limestone
$MgCO_3$	Magnesite
$CaCO_3 \cdot MgCO_3$	Dolomite
$FeCO_3$	Siderite
$ZnCO_3$	Calamine
$Cu(OH)_2 \cdot CuCO_3 / Cu_2(OH)_2CO_3$	Malachite or Basic copper carbonate
$Cu(OH)_2 \cdot 2CuCO_3$	Azurite
$PbCO_3$	Cerrusite

b. Sulphate ores

$CaSO_4 \cdot 2H_2O$	Gypsum	$K_2SO_4 \cdot MgSO_4 \cdot MgCl_2 \cdot 6H_2O$	Kainite
$MgSO_4 \cdot 7H_2O$	Epsom salt	$MgSO_4 \cdot H_2O$	Kieserite
$PbSO_4$	Anglesite	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 4Al(OH)_3$	Alunite
$BaSO_4$	Baryte	$PbO \cdot PbSO_4$	Lanarkite
$Na_2SO_4 \cdot 10H_2O$	Glauber Salt		
$CuSO_4 \cdot 5H_2O$	Chalcanthite		

c. Nitrate ores

KNO_3	Indian saltpeter
$NaNO_3$	Chile saltpeter